4.1 Complex Numbers

Name _____ Period ____ Date _____

Pre-Calculus 4.1 (30-minute plan) · Student sheet

I can…

- write square roots of negative numbers using i.
- add, subtract, and multiply complex numbers.
- divide complex numbers using the complex conjugate.

Vocabulary · fill in the missing word

Imaginary unit i — i = _____, so $i^2 = -1$

Complex number — a number a + bi, where a is the real part and b is the _____ part

Complex conjugates — a _____ of complex numbers a + bi and a − bi

Principal square root of a negative number — $\sqrt{-a}$ = _____, for a > 0

Part 1 · Notes and examples (fill in each note, then work the example)

The imaginary unit: i = $\sqrt{-1}$, so i^2 = _____.

Adding and subtracting: combine the _____ parts and the imaginary parts.

Example 1

a. (3 − i) + (2 + 3i)

b. (4 + 2i) − (7 − 5i)

Multiplying: use FOIL, then replace i^2 with _____.

Example 2

a. (2 − i)(4 + 3i)

b. (3 + 2i)(3 − 2i)

Dividing: multiply the numerator and denominator by the _____ of the denominator.

Square roots of negatives: write $\sqrt{-a}$ as _____ BEFORE multiplying.

Example 3

Write in standard form: $\frac{2 + 3i}{4 - 2i}$

Part 2 · Partner practice

Partner practice

Write in standard form: $\frac{3 + i}{1 - 2i}$